

## Lab – Agents with Agno

### **Ex1 – Simple Agent**

Create an Agent for summarizing general information of any topic.

Use role, goal, description and preferred model. The summary will be formed of 7 bullet list

### **Ex2 – Agent with Tools**

Create an Agent that show ONLY a name, user name and email of a user by his given id.

The agent used a tool that receives a user id and pulled his data from the external api : [https://jsonplaceholder.typicode.com/users/\[userid\]](https://jsonplaceholder.typicode.com/users/[userid]),

For example : <https://jsonplaceholder.typicode.com/users/4>

The tool returned the whole user data, but the agent show only the requested data (name, email and user name)

### Ex3 – Agents with Memory and RAG

- Create an Agent that
  - o Uses a tool that receives “randomized” weather conditions from the options : [“Cloudy”, “Rainy”, “Snowy”, “Windy”]
  - o Uses knowledgebase (about climate changes) load from <https://uxplanet.org/10-things-you-should-know-about-your-users-478dd73a1356>
  - o Uses a long term storage

The Agent prefer using data from the memory if exists. If not, from the knowledge base and only then using the tool (as last option)

### Ex4 – Agents with User Inputs

You need to create an agent that calculate a product price after a given discount. The agent use 2 tools

- One for returning a product price for a given product name taken from <https://fakestoreapi.com/products>
- One for evaluating the price after the discount, using Python built-in “eval” function

If the product name and the discount percentage are not exist, they must be asked from the user. that means that the agent must supports BOTH following prompts:

- What is the final price of ‘**Opna Women's Short Sleeve Moisture**’ after a **20%** discount?
- What is the final price after a discount?

## Ex5 – Team Of Agents

Build a Team of Agents for running Python code.

- The first agent create the code for a given task (will be given in the prompt)
- The second Agent running that code (using the **agno.tools.python.PythonTools()** tool)
- The third agent document each step in the code

An example of a given task to the team:

**A function that sorts the following list of numbers : [5,2,8,1]**

## Final Ex

You need to create a Team of Agents for planning a trip to a given city.  
The Team composed from the following Agents:

- **Info Agent**

An agent that pulled general information about a city using  
DuckDuckGoTools tool

- **Weather Forecast Agent**

An Agent that forecasts weather for the next days in a given city  
That agent uses a tool you must build.

That tool gets a city and number of days and uses the following api for  
getting that data :

`http://api.weatherapi.com/v1/forecast.json?key=09587f210a5c48d5bae19235825  
2202&q={city}&days={num_of_days}`

The relevant returned data is located in the  
response["forecast"]["forecastday"] key .

Each day forecast (in that list) will be as follows :

```
{
  'date' : date["date"],
  'max_temp' : date["day"]["maxtemp_c"] ,
  'condition' : date["day"]["condition"]["text"]
}
```

- **Recommendations Agent**

An agent that returned recommendations for top 5 attractions in that city.  
That agent does NOT use any tools

Important : Give the agents and the Team proper instructions !

Use that Team of Agents with the following prompt :  
“Plan a trip to London for the next 3 days”

**Extras :**

1. The Info Agent brings data about that city if there is no stored data already (use storage)
2. Add another agent for gathering demographic data for that city using a MCP Server (The exist MCP Server is located in the Lab folder)